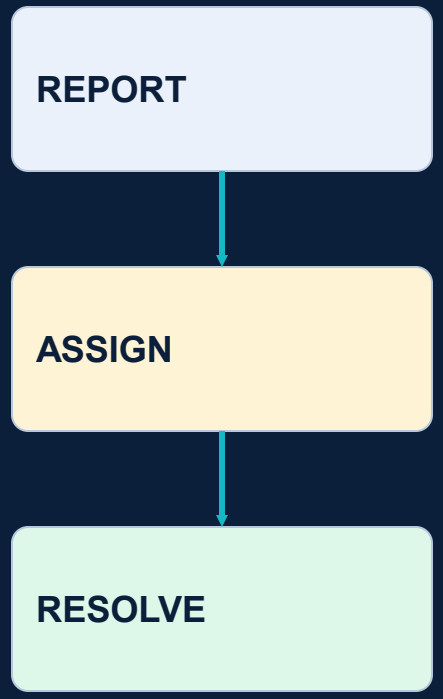
CampusFix

A Role-Based College Complaint and Maintenance Management System

Fourth-Year CSE Mini Project | Academic Session 2026-27

[Student Name] • [Roll Number]
Guide: [Faculty Guide Name]

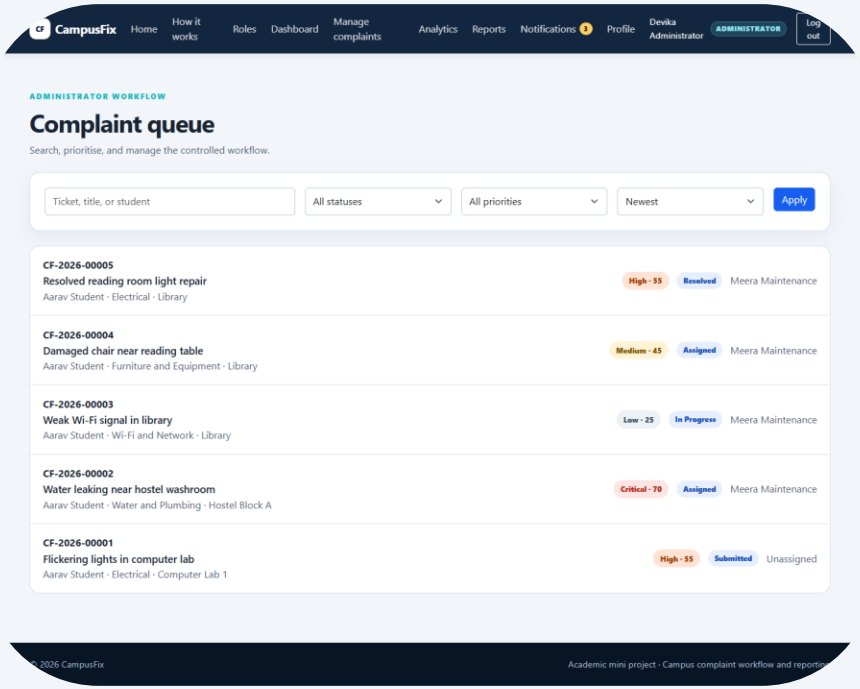


Problem Statement

Informal reporting loses ownership, status, and evidence

A maintenance issue may be reported quickly—then become difficult to trace.

- No unique ticket or shared source of truth
- Unclear assignment and responsibility
- No permanent record of status changes
- Students cannot see progress
- Administrators lack workload evidence



Why CampusFix

One shared ticket from report to verified completion

1

Traceability

Unique ticket and permanent timeline

2

Responsibility

Role-specific assignment and actions

3

Visibility

Notifications, analytics, and reports

Outcome: every accepted complaint has an owner, a state, and an inspectable history.

Project Objectives

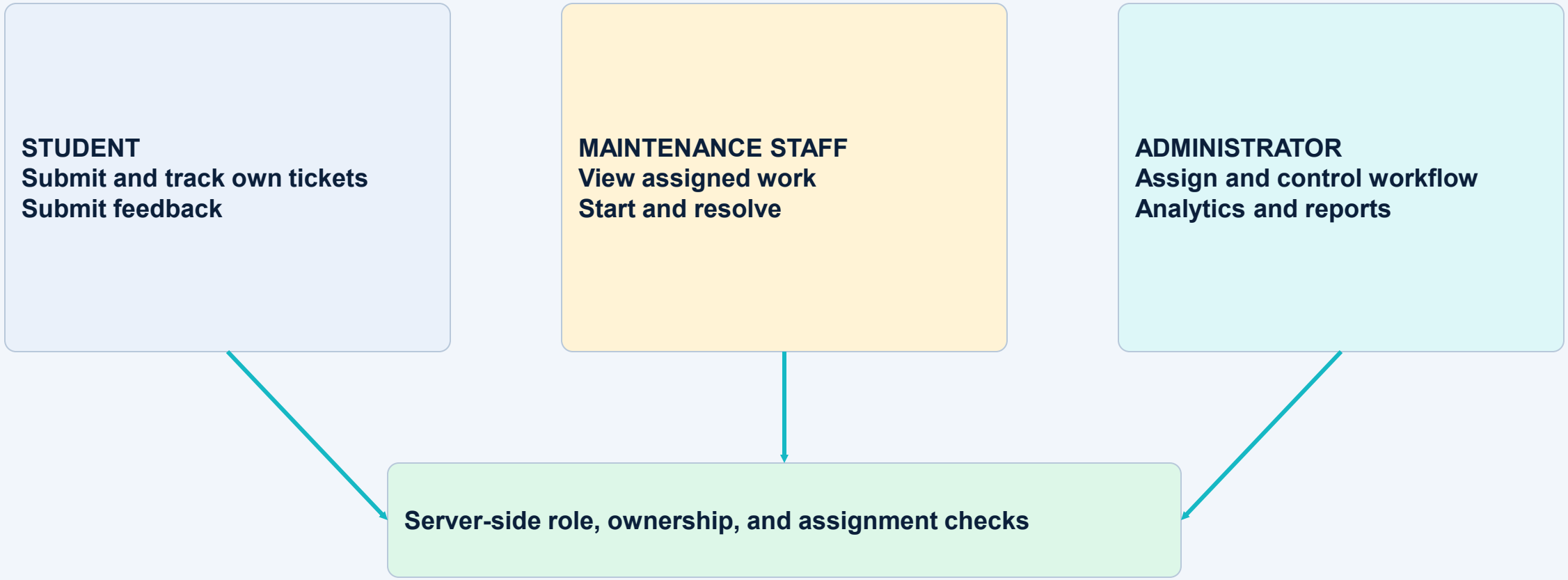
- Centralise campus maintenance complaints
- Enforce Student, Staff, and Administrator permissions
- Control the complaint lifecycle through valid transitions
- Preserve permanent workflow history
- Provide transparent priority and private notifications
- Measure outcomes through feedback, analytics, and reports
- Verify behaviour with automated and live HTTP tests

Existing Problems and CampusFix Solution

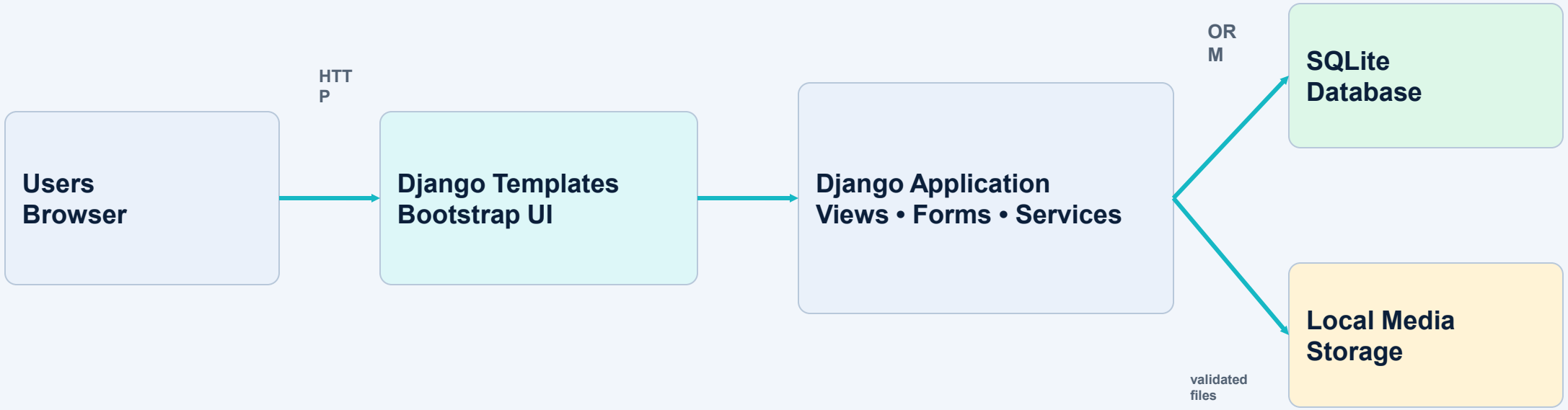
Existing approach		CampusFix
Verbal or scattered messages	→	**One searchable ticket record**
No clear owner	→	**Controlled assignment to active Staff**
Status known only by asking	→	**Live status, timeline, notifications**
Subjective urgency	→	**Explainable priority score**
No review evidence	→	**Analytics, feedback, CSV, print reports**

User Roles

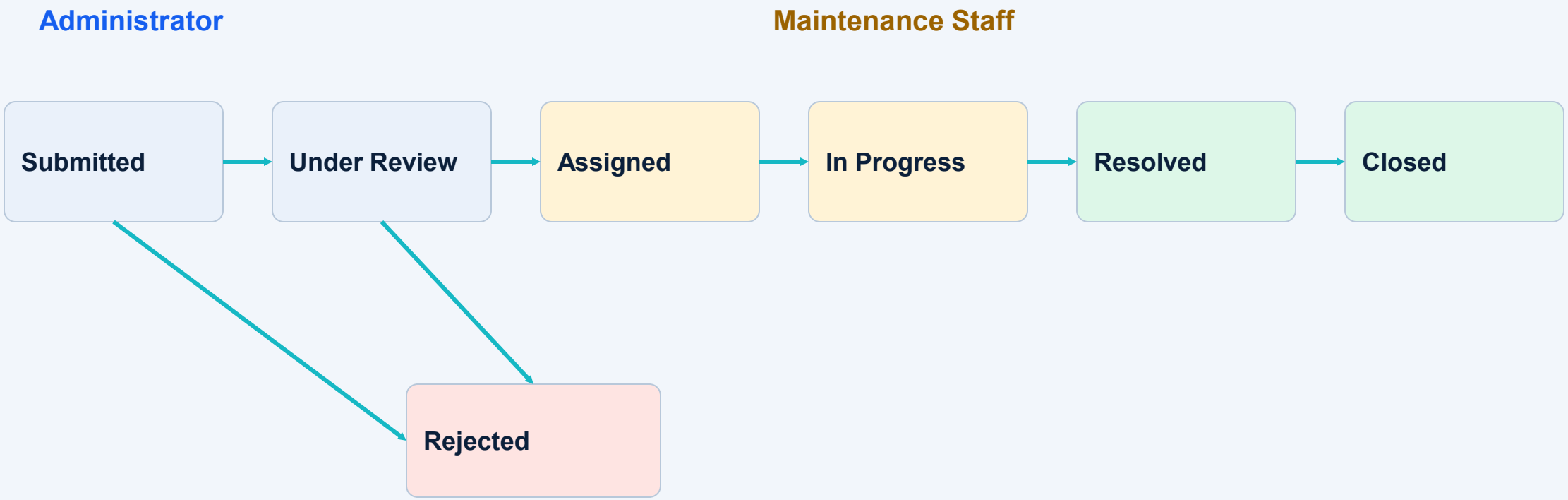
Permissions are enforced on the server



System Architecture



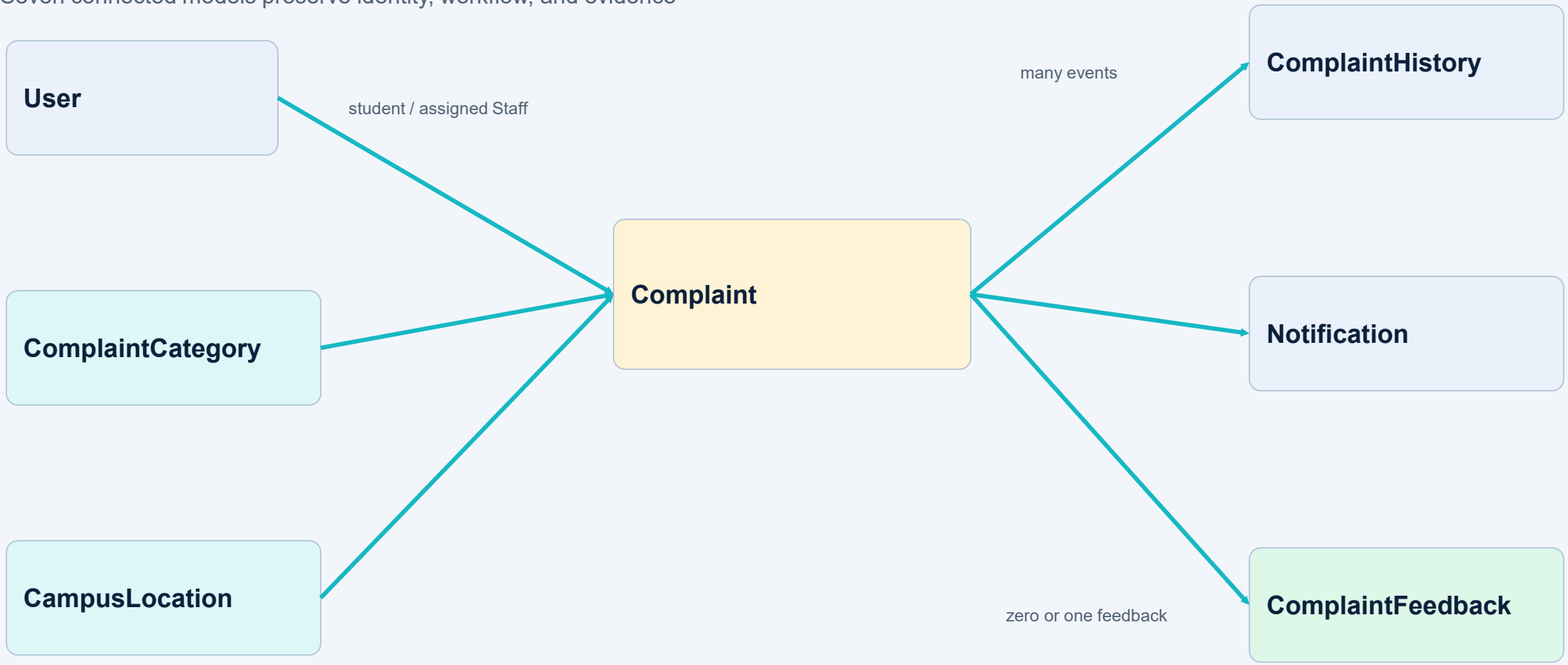
Complaint Lifecycle and Role Workflow



Every successful transition appends a permanent history event.

Database Design

Seven connected models preserve identity, workflow, and evidence



Technology Stack

Python

Application language

Django 6.1.1

Auth • ORM • forms • tests

SQLite

Local transactional database

Bootstrap 5.3.8

Responsive interface

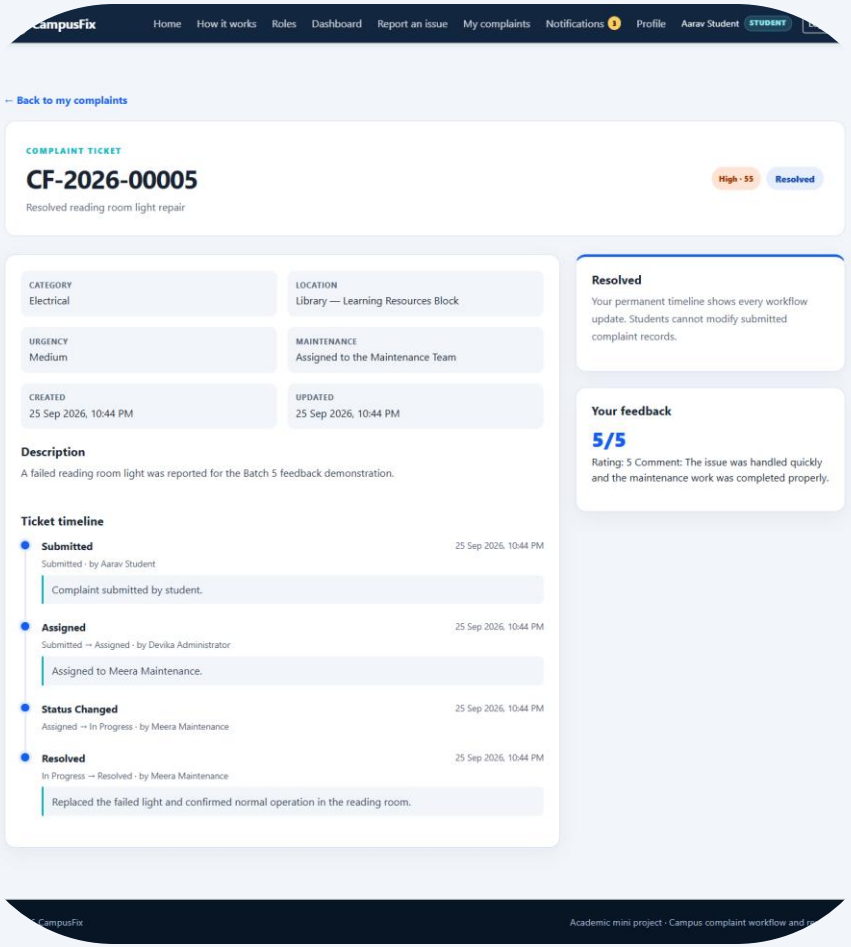
Chart.js 4.4.7

Administrator charts

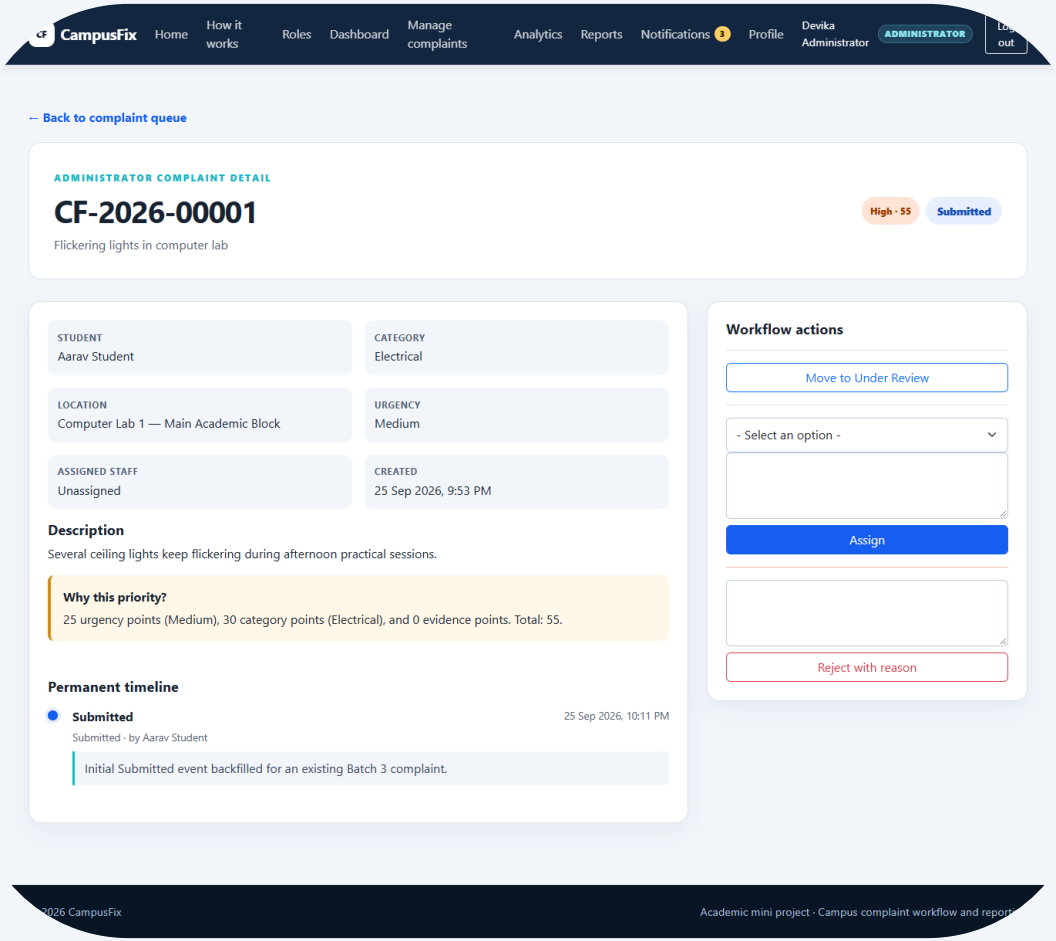
Pillow 11.3.0

Image content validation

Major Features

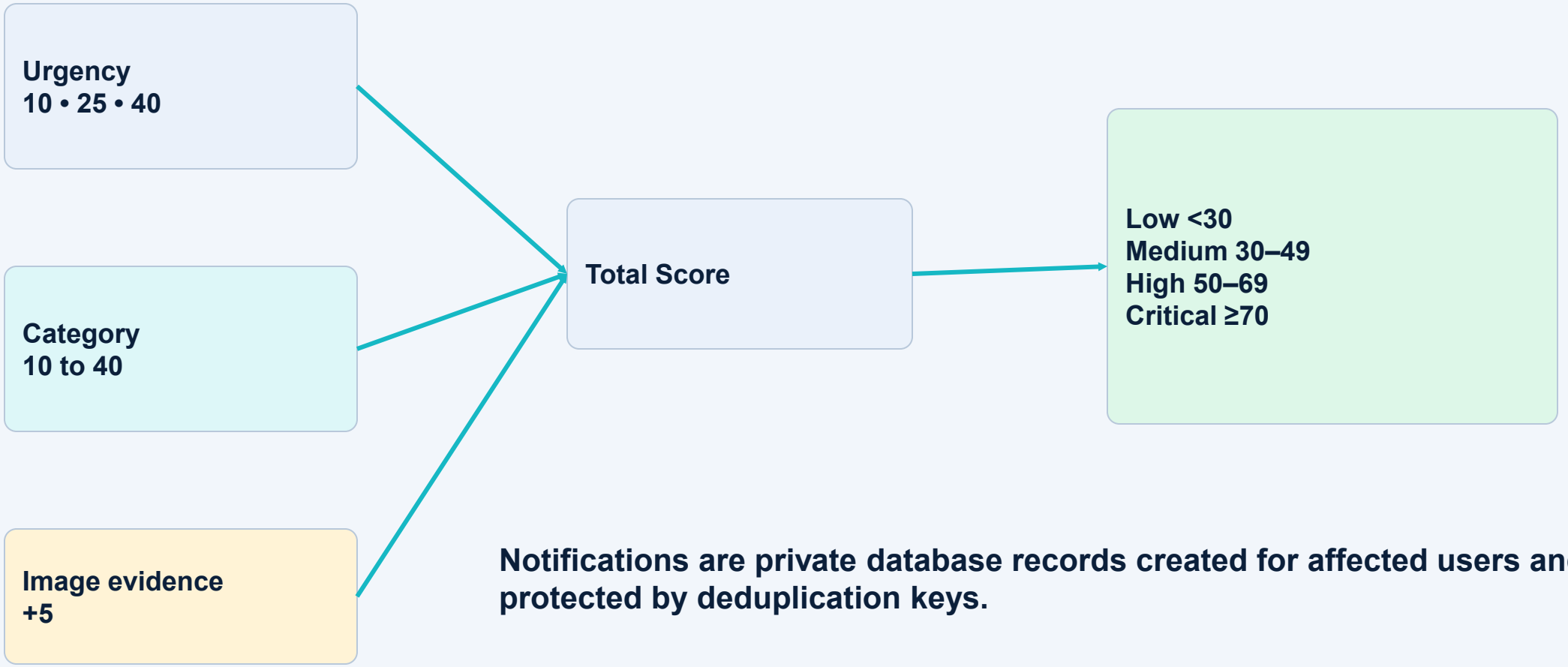


Student evidence and timeline



Administrator controls and audit trail

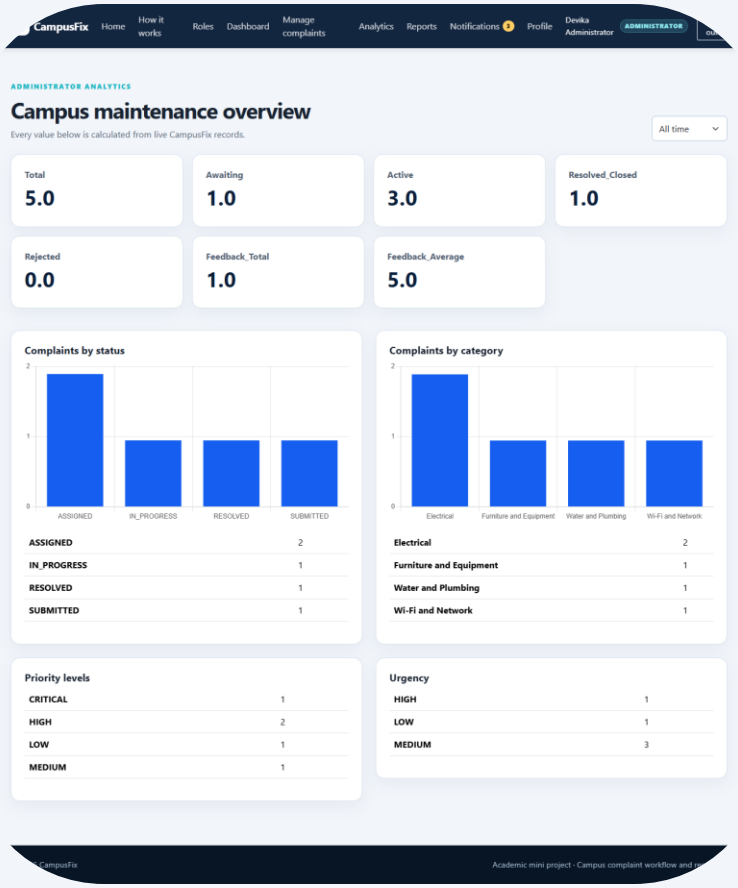
Priority Scoring and Notifications



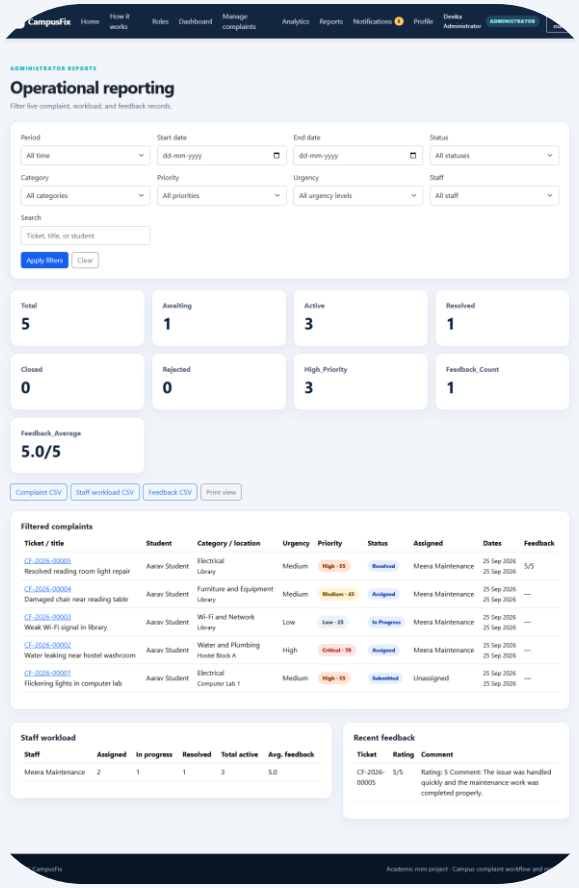
Notifications are private database records created for affected users and protected by deduplication keys.

Analytics and Reporting

Live database records become operational evidence

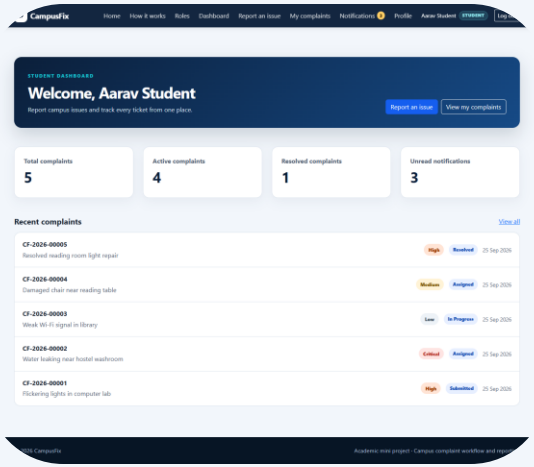


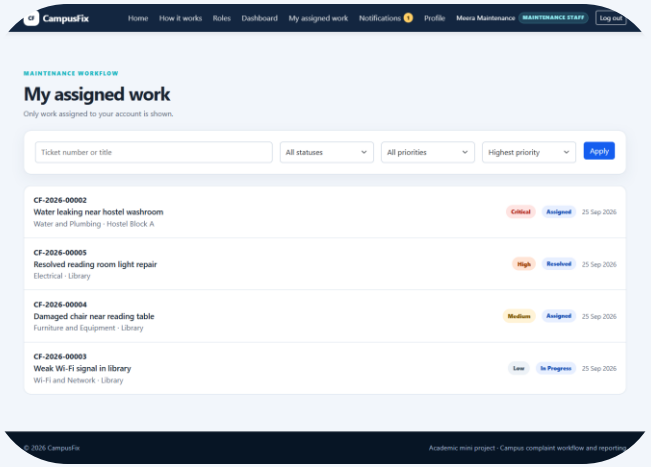
Charts plus accessible tables

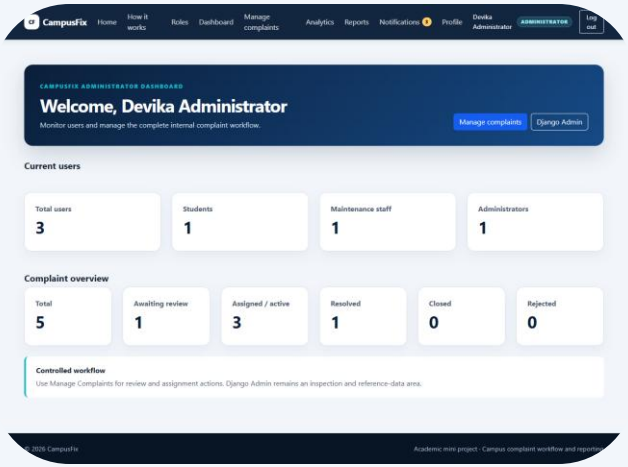


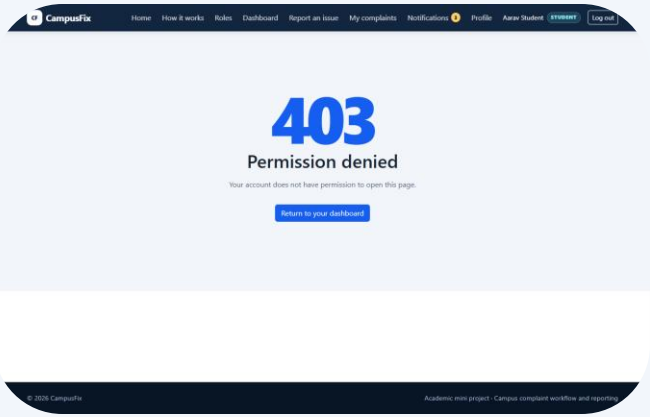
Filters • workload • feedback • CSV • print

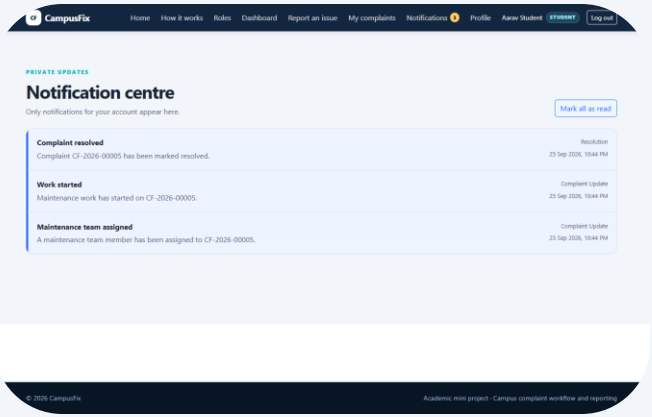
Working Demonstration Evidence











Testing and Results

End-of-Batch-6 verification

84
tests passed

0
system-check issues

0
missing migrations

200
admin report HTTP

403
student / staff reports

- Model, form, page, ownership, and permission tests
- Controlled workflow and permanent history tests
- Notification, feedback, analytics, and reporting tests
- CSV headers, filters, UTF-8, and formula-safety tests
- Live no-reloader server smoke checks

Conclusion and Future Scope

COMPLETED IMPLEMENTATION

- Role-based end-to-end complaint workflow
- Permanent timeline and private notifications
- Explainable priority, feedback, analytics, and reports
- Responsive interface and verified security boundaries

FUTURE SCOPE

- Email and SMS notifications
- Mobile or progressive web application
- PostgreSQL, cloud media, HTTPS, backups, and monitoring
- Predictive maintenance and richer service analytics
- Multilingual interface

Current implementation remains local and intentionally stops before production deployment.

Thank You

Questions and Discussion



CampusFix

[Student Name] • [Roll Number]